



THE UNIVERSITY OF
SYDNEY

Room Number	
Seat Number	
Student Number	

ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Computer Science

EXAMINATION

Semester 1 - Final, 2025

COMP4349/COMP5349 Cloud Computing

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:

This is a CLOSED book exam - no material permitted.

MATERIALS PERMITTED IN THE EXAM VENUE:

(No electronic aids are permitted e.g. laptops, phones)

Blank scratch paper

MATERIALS TO BE SUPPLIED TO STUDENTS:

INSTRUCTIONS TO STUDENTS:

This exam consists of nine short answer questions worth a total of 100 marks.

Answer all questions in the spaces provided on this paper.

You may use pencil or ink.

If you need additional writing space, please use the extra pages provided at the end of this exam booklet.

Only pages in this exam booklet will be marked.

Please tick the box to confirm that your examination paper is complete. ☐

Disclaimer

This is the final exam script used in 2025. The questions are based on 2025 syllabus, which covers slightly different topics and has different assignments. The exam script is released as an example of the short answer question format. The solutions will not be released. Tutors may discuss certain questions during the lab if time permits.

Question 1 [6 points] What is the AWS Nitro system and how does it improve virtualization? List two key features of Nitro System to answer this question.

Question 2 [6 points] Compare S3 redundancy and versioning. What is the key difference in their purpose and behavior?

Question 3 [6 points] Describe the difference between Amazon RDS Multi-AZ deployment and Read Replica.

Question 4 [8 points] Compare the architectural considerations and operational aspects of implementing microservices using containers and serverless computing. For each implementation method, provide a detailed example of a use case where its strengths would be most beneficial.

Question 5 [8 points] You are given the following IAM policy, which is attached to a user in your AWS account:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Deny",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "203.0.113.0/24"
        }
      }
    }
  ]
}
```

Answer the following questions

1. **[2 points]** What EC2 actions are affected by this policy?

2. **[2 points]** What will happen if the user attempts to stop an EC2 instance from IP address 203.0.113.17? Explain your answer.

3. [2 points] Would the user be able to start instances from IP 198.51.100.10? Explain your answer.

4. [2 points] Modify the policy so that the **deny applies to all IPs except** 198.73.100.0/24.

Question 6 [15 points] Your team is working on an archival storage solution using Amazon S3. The bucket `project` has versioning enabled, and it is configured to replicate to a destination bucket (`project-backup`) in a different region with default setting. Over the past few days, the following actions were performed on an object named `report.csv` in that bucket `project` :

1. On Day 1, `report.csv` is uploaded (Version A).
 2. On Day 2, a new version of `report.csv` is uploaded (Version B).
 3. On Day 3, `report.csv` is deleted via the AWS Console.
1. **[3 points]** How many versions of `report.csv` exist in the bucket `project` after Day 3? List all versions

2. **[3 points]** Can you recover `report.csv` after the deletion on Day 3 in bucket `project`? If so, how?

3. [3 points] How many versions of `report.csv` exist in the bucket `project-backup` after Day 3? List all versions

4. [3 points] What will happen if you attempt to download `report.csv` using the default (unversioned) S3 GET request after Day 3 from bucket `project`?

5. [3 points] How would you **permanently delete** all versions of `report.csv` from both buckets?

Question 7 [12 points] Amazon Aurora is a cloud-native relational database service that separates the compute layer (database engine) from its distributed, fault-tolerant storage layer. Your application is using Amazon Aurora MySQL with **one writer instance and two reader replicas**. Your database volume is currently stored in one protection group.

1. **[2 points]** How many copies of your data are stored by Aurora's storage layer? Explain your answer.

2. **[2 points]** Across how many Availability Zones is your data stored? Explain your answer

3. [4 points] When your application writes data to Aurora, describe how the write is propagated to various nodes.

4. [4 points] If your database volume grows beyond the capacity of a single protection group, which additional resources are needed to support your database.

Question 8 [30 points]

Your company has deployed a web application in a custom VPC (10.0.0.0/16). The deployment architecture includes:

Subnets and Resources

Subnet Name	CIDR	Associated Route Table	Resources
Public Subnet	10.0.1.0/24	RT-Public	ALB, NAT Gateway, Bastion Host
App Subnet	10.0.2.0/24	RT-App	Web servers (EC2)
DB Subnet	10.0.3.0/24	Default Route Table	RDS MySQL Database

Route Tables

- **RT-Public:**
 - 0.0.0.0/0 → **Internet Gateway**
- **RT-App:**
 - 0.0.0.0/0 → **NAT Gateway**
- **Default Route Table**
 - 10.0.0.0/16 → **local**

Security Groups

- **ALB SG:**
 - Inbound: TCP 80 from 0.0.0.0/0
- **Bastion SG:**
 - Inbound: TCP 22 from 0.0.0.0/0
- **Web Server SG:**
 - Inbound: TCP 80 from ALB SG
- **RDS SG:**
 - Inbound: TCP 3306 from Web Server SG

1. **[2 points]** Is the RDS able to connect to the internet? Explain your answer.

2. [4 points] Describe the complete network path when a user accesses the web application from the internet. What route tables and security groups are involved?

3. [4 points] Is the Bastion Host useful in this setting? If yes, explain how; If not, how can you update various configuration to make it useful.

4. [4 points] A developer adds an EC2 instance in the DB subnet and assigns it a public IP. They are still unable to SSH into it from their laptop. Explain two reasons why this setup fails.

5. [6 points] You are designing a lambda function that needs to access both the RDS and S3. You decide to associate the lambda function with a private subnet. Describe which subnet you should put it in and what other configurations are needed to ensure that the lambda function can interact with both.

6. [4 points] A developer needs to temporarily launch an EC2 instance to connect to the RDS instance using MySQL client. Which subnet should they choose for the EC2 instance. Explain your choice based on network configuration. Describe how can the developer connect to the instance to run MySQL client. If new SG needs to be created explain that.

7. [6 points] For each of the following IP addresses, state which subnet it belongs to and what that implies for its routing and external accessibility:

- (a) 10.0.2.15

- (b) 10.0.3.10

- (c) 10.0.5.55

Question 9 [9 points]

Your company runs a web application on AWS. You've deployed the application on Amazon EC2 instances behind an **Application Load Balancer (ALB)**, and you've configured an **Auto Scaling Group (ASG)** to automatically adjust the number of instances based on CPU utilization.

The setup includes:

- The **ALB** listens on port 80 and forwards traffic to a **Target Group** associated with the ASG.
- The **ALB** periodically check the health of **Target Group** on port 80 of **EC2 instance**.
- The **ASG** launches EC2 instances using a Launch Template with a startup script that installs and starts a Python web server on port 80.
- **Scale-in and scale-out policies** are triggered based on CPU usage, with a policy to:
 - **Scale out when average CPU > 70%**
 - **Scale in when average CPU < 30%**
- The **minimum size** of the ASG is 2, and the **maximum size** is 5.

1. **[2 points]** If the ASG group currently has 3 instances, how would ALB distribute traffic among this three?

2. [**4 points**] Assume you made some update to the application and created a new version of the launch template and configured your ASG to use the new version. Unfortunately, the startup script in the launch template has a bug and fails to start the web server. What would happen to your ASG? What observation may trigger you to check the script?

3. [**3 points**] If the ASG group currently has 2 instances running. One instance has CPU at **90%**, and the other at **10%**. Will the ASG scale out in this situation? Why or why not?

Additional Answer Space

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